22

Duality and Bilinear Forms

Historical Note

The transpose is old: Cauchy was using it in 1815 in his work on determinants, and for a century it was simply an operation on arrays, defined by swapping the indices and justified by the fact that it worked. What it *is* took much longer to isolate. The answer, that a linear map induces a map running the other way on the functionals, belongs to the theory of duality that Grassmann anticipated in 1844, that Ricci and Levi-Civita encoded around 1900 in the distinction between covariant and contravariant indices, and that Banach named and made systematic in 1932 for spaces of infinite dimension. The finite-dimensional case is the easy one, and this chapter is about it. The second half, the classification of bilinear forms, is older still: Gauss reduced quadratic forms in the *Disquisitiones* of 1801, and Jacobi and Sylvester finished the story in the 1850s.

Two loose ends motivate this chapter. Chapter 7 observed that textbooks disagree about whether a matrix or its transpose deserves to be called the matrix of a map, and did not settle it. Chapter 15 introduced *congruence*, $A \mapsto C^{t}AC$, as a change of variables in a quadratic form, without saying why that rule differs from the similarity $A \mapsto C^{-1}AC$ of Chapter 7. Both are answered by the same observation: a matrix can represent two quite different things, and which one it represents decides how it transforms.

22.1 Linear functionals and the dual space

Definition 22.1 (Dual space)**.** *Let V be a vector space over $\mathbb{F}$. A* linear functional *on V is a linear map $V \to \mathbb{F}$. The set of them is the* dual space

$$V^{*} \;=\; \mathrm{Hom}(V, \mathbb{F}),$$

a vector space under pointwise operations by Proposition 7.7.

Example 22.2 (Functionals one already knows). *On $\mathbb{R}^n$, the map $x \mapsto \langle x, w\rangle$ for a fixed w; on $M_n(\mathbb{R})$, the trace; on polynomials, evaluation $p \mapsto p(3)$, or the integral $p \mapsto \int_0^1 p$; on the space of solutions of a differential equation, the value at a chosen time. Each is a rule that eats a vector and returns one number, linearly.*

Proposition 22.3 (The dual basis). *Let $v_1, \dots, v_n$ be a basis of V. For each i let $f_i \in V^*$ be the functional with*

$$f_i(v_j) \;=\; \delta_{ij} \;=\; \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

Then $f_1, \dots, f_n$ is a basis of V^*, so $\dim V^* = \dim V$. Moreover every $v \in V$ satisfies $v = \sum_i f_i(v)\, v_i$: the dual basis reads off coordinates.

Proof. Each f_i exists and is unique by Theorem 7.5, since a linear map is a free choice of values on a basis. For spanning, let $g \in V^*$ and put $h = \sum_i g(v_i) f_i$. Then $h(v_j) = g(v_j)$ for every j, so $h = g$ by Theorem 7.5 again. For independence, if $\sum_i c_i f_i = 0$ then evaluating at v_j gives $c_j = 0$. The last claim is the spanning computation applied to the identity map, or directly: writing $v = \sum_j a_j v_j$ gives $f_i(v) = a_i$. □

Figure 22.1 draws a dual basis vector the way a functional should be drawn, as the family of hyperplanes on which it is constant. Doubling the functional halves the spacing, so the size of a functional is how tightly its level sets are packed.

Caution

Proposition 22.3 gives an isomorphism $V \to V^*$, sending $v_i \mapsto f_i$, but it depends on the basis and changes when the basis changes. There is no *canonical* isomorphism between a space and its dual, and the two should be thought of as different spaces that happen to have the same dimension. What does exist canonically is the identification of V with its double dual, in Theorem 22.4, and an identification of V with V^* once an inner product has been chosen, in Proposition 22.5. An inner product is precisely the extra datum that makes rows and columns interchangeable, which is why so much of Chapters 12 to 15 could quietly treat them as the same.

Theorem 22.4 (The double dual). *For finite-dimensional V the evaluation map*

$$\mathrm{ev} : V \to V^{**}, \qquad \mathrm{ev}(v)(f) = f(v),$$

is an isomorphism, and it is defined without reference to any basis.

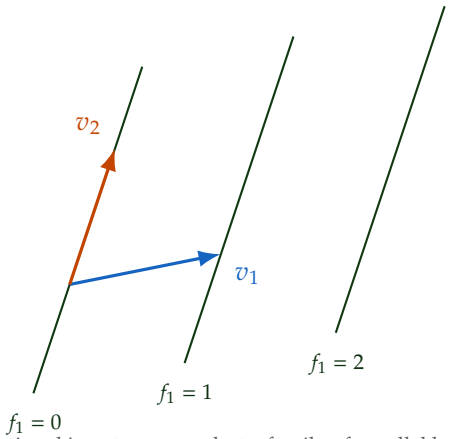


Figure 22.1. Proposition 22.3. The dual basis vector f_1 is drawn the way a functional should be drawn, as the family of hyperplanes on which it is constant. Doubling f_1 halves the spacing, so the "size" of a functional is how tightly its level sets are packed. This picture is also the reason a gradient is not really a vector: it is the object that assigns a number to a step, which is exactly a functional.

Proof. $\mathrm{ev}(v)$ is linear in f, so lands in V^{**}, and ev is linear in v. It is injective: if $v \neq \mathbf{0}$, extend v to a basis and let f be the first dual basis vector, so $f(v) = 1$ and $\mathrm{ev}(v) \neq 0$. Since $\dim V^{**} = \dim V^{*} = \dim V$ by Proposition 22.3 applied twice, an injective map between spaces of equal finite dimension is an isomorphism by Corollary 7.13. □

Proposition 22.5 (Riesz representation, finite dimensions)**.** *Let V carry an inner product. Then $w \mapsto \langle \cdot, w \rangle$ is an isomorphism $V \to V^{*}$ in the real case, and a conjugate-linear bijection in the complex case. Every functional is $\langle \cdot, w \rangle$ for exactly one w.*

Proof. Take an orthonormal basis $u_1, \dots, u_n$, which exists by Theorem 12.10. Given $g \in V^{*}$, put $w = \sum_i \overline{g(u_i)}\, u_i$. Then $\langle u_j, w \rangle = g(u_j)$ for every j, so $\langle \cdot, w \rangle = g$. Uniqueness holds because $\langle \cdot, w \rangle = \langle \cdot, w' \rangle$ makes $w - w'$ orthogonal to everything, hence zero. □

Key Idea

Proposition 19.1 of Chapter 19, which said that every linear score is a dot product, was this theorem in the special case $V = \mathbb{R}^d$. It is worth knowing that it has a name and that the inner product is doing the work: without one, a functional is not a vector and cannot be compared with the vectors it eats.

22.2 The dual map, and what a transpose is

Definition 22.6 (Dual map). *For linear $T : V \to W$, the* dual map $T^t : W^* \to V^*$ *is*

$$T^t(g) \;=\; g \circ T.$$

It sends a functional on the target to a functional on the source, by measuring after the map instead of before.

Figure 22.2 shows what that does to a diagram: dualising reverses every arrow. That reversal is the whole reason a transpose reverses products.

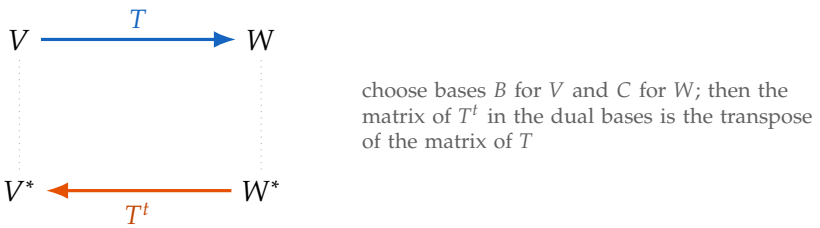


Figure 22.2. Definition 22.6 and Theorem 22.7. Dualising reverses every arrow, because a functional on W is turned into one on V by composing with T on the right. That reversal is the whole reason transposes reverse products, and therefore the reason the backward pass of Chapter 19 runs backwards: the gradient is a functional, and functionals travel against the maps.

Theorem 22.7 (The transpose is the matrix of the dual map). *Let $B = (v_1, \dots, v_n)$ be a basis of V with dual basis B^*, and $C = (w_1, \dots, w_m)$ a basis of W with dual basis C^*. Then*

$$\big[T^t\big]_{B^* \leftarrow C^*} \;=\; \big([T]_{C \leftarrow B}\big)^t.$$

Proof. Write $A = [T]_{C \leftarrow B}$, so that $Tv_j = \sum_i a_{ij} w_i$ by Definition 7.17. Let $g_1, \dots, g_m$ be the dual basis C^* and $f_1, \dots, f_n$ the dual basis B^*. Then

$$\big(T^t g_i\big)(v_j) \;=\; g_i(Tv_j) \;=\; g_i\Big(\sum_k a_{kj} w_k\Big) \;=\; a_{ij},$$

so by Proposition 22.3 the functional $T^t g_i$ has coordinates $(a_{i1}, \dots, a_{in})$ in the basis B^*, that is $T^t g_i = \sum_j a_{ij} f_j$. Those coordinates form the ith *column* of the matrix of T^t and the ith *row* of A, which is the assertion. □

Key Idea

So the disagreement flagged at the end of Chapter 7 is not a disagreement about convention. A matrix and its transpose are the matrices of two different maps, between two different pairs of spaces, and the reason both get called "the matrix of T" is that an inner product lets one forget the difference. Everything the book has proved about transposes now has a one-line reason:

$$(ST)^t = T^t S^t \quad \text{because} \quad g \circ (S \circ T) = (g \circ S) \circ T,$$

and $\mathbf{1}^t = \mathbf{1}$, and T invertible forces T^t invertible with $(T^{-1})^t = (T^t)^{-1}$. Composition of pullbacks reverses the order, and the matrix identity of Chapter 4 is its shadow.

Caution

The dual map T^t and the adjoint T^* of Chapter 14 are not the same construction, though the book's notation makes them look alike. The dual map exists for every linear map between any vector spaces and needs no extra structure; the adjoint needs an inner product on both sides, and is defined by $\langle Tv, w\rangle = \langle v, T^*w\rangle$. With inner products chosen, Proposition 22.5 identifies V^* with V and W^* with W, and under that identification the dual map becomes the adjoint. In orthonormal bases the two therefore have the same matrix, up to the conjugation that Definition 12.16 inserts in the complex case. In non-orthonormal bases they differ.

22.3 Annihilators, and rank one more time

Definition 22.8 (Annihilator)**.** *For a subspace $U \subseteq V$, its* annihilator *is $U^\circ = \{f \in V^* : f(u) = 0$ for every $u \in U\}$, a subspace of V^*.*

Proposition 22.9 (Dimension of an annihilator)**.** $\dim U + \dim U^\circ = \dim V$.

Proof. Take a basis $v_1, \ldots, v_k$ of U and extend it to a basis $v_1, \ldots, v_n$ of V by Theorem 6.14. A functional vanishes on U exactly when it vanishes on $v_1, \ldots, v_k$, that is exactly when its expansion in the dual basis uses only $f_{k+1}, \ldots, f_n$. So U° has $f_{k+1}, \ldots, f_n$ as a basis and dimension $n - k$. □

Theorem 22.10 (Kernel and image, dualised)**.** *For $T : V \to W$,*

$$\ker\big(T^t\big) = \big(\operatorname{Im} T\big)^\circ, \qquad \operatorname{Im}\big(T^t\big) = \big(\ker T\big)^\circ.$$

Consequently $\operatorname{rank} T^t = \operatorname{rank} T$.

Proof. For the first, $g \in \ker T^t$ means $g(Tv) = 0$ for every v, which is exactly that g annihilates $\operatorname{Im} T$.

For the second, if $g \in W^*$ and $v \in \ker T$ then $(T^t g)(v) = g(Tv) = 0$, so $\operatorname{Im} T^t \subseteq (\ker T)^\circ$. The two have the same dimension: by rank and nullity, Theorem 7.10, and Proposition 22.9,

$$\dim(\ker T)^\circ = \dim V - \dim \ker T = \operatorname{rank} T,$$

while by the first part and Proposition 22.9 again,

$$\operatorname{rank} T^t = \dim W^* - \dim \ker T^t = \dim W - \dim(\operatorname{Im} T)^\circ = \dim \operatorname{Im} T = \operatorname{rank} T.$$

So the inclusion is an equality, and the rank statement has been proved along the way. □

Insight

That is the third proof in this book that a matrix and its transpose have the same rank. The first, in Chapter 6, counted pivots; the second, in Chapter 16, observed that A and A^t have the same singular values; this one uses no computation and no geometry at all. It also puts the four fundamental subspaces of Theorem 13.12 in their proper setting. There, $N(A^t)$ was the orthogonal complement of $C(A)$ inside the same $\mathbb{R}^m$; here it is the annihilator, which lives in the dual space rather than alongside the original. The orthogonal complement is what the annihilator becomes once an inner product is available to drag it back, and the theorem holds over any field, where no inner product exists.

22.4 Bilinear forms

Definition 22.11 (Bilinear form and its matrix)**.** *A* bilinear form *on V is a map $\beta : V \times V \to \mathbb{F}$ linear in each argument separately. Given a basis $v_1, \dots, v_n$, its* matrix *is A with $a_{ij} = \beta(v_i, v_j)$, and then*

$$\beta(x, y) \;=\; [x]^t A\, [y]$$

for coordinate vectors $[x]$, $[y]$.

Proposition 22.12 (Change of basis for a form)**.** *If C is the change of basis matrix, the matrix of β becomes C^tAC. Two matrices related this way with C invertible are* congruent*, as in Chapter 15.*

Proof. Write $[x]$ for the new coordinates, so that the old ones are $C[x]$. Then

$$\beta(x,y) \;=\; (C[x])^t A\,(C[y]) \;=\; [x]^t\,(C^tAC)[y],$$

which is the assertion. □

So a change of basis acts on a form by $A \mapsto C^tAC$ and on an operator by $A \mapsto C^{-1}AC$, and Figure 22.3 puts the two rules side by side. They agree exactly when C is orthogonal, which is why the distinction can be missed for a long time.

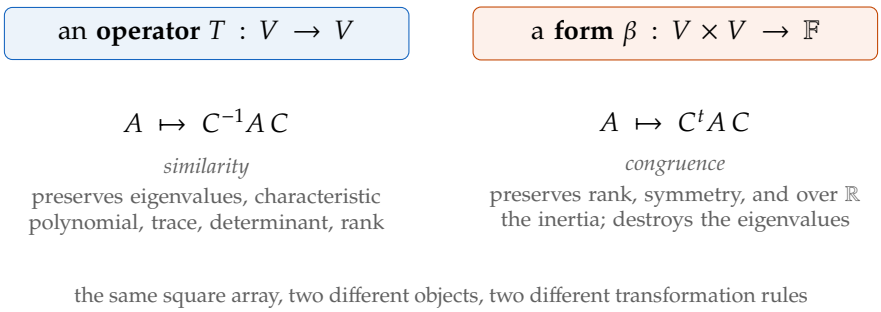


Figure 22.3. Proposition 22.12 against Chapter 7. A matrix is not a thing in itself, and which rule applies depends on what it is the matrix *of*. For a symmetric A the two rules agree exactly when C is orthogonal, since then $C^{-1} = C^t$, which is why the spectral theorem of Chapter 14 can diagonalise a symmetric matrix as an operator and as a form at the same time, and why Theorem 15.10 is the best that congruence alone can do.

Proposition 22.13 (Symmetric plus alternating)**.** *Over $\mathbb{R}$ or $\mathbb{C}$, every bilinear form is uniquely the sum of a* symmetric *form, with $\beta(x,y) = \beta(y,x)$, and an* alternating *one, with $\beta(x,x) = 0$ for every x. In matrices, $A = \frac{1}{2}(A + A^t) + \frac{1}{2}(A - A^t)$.*

Proof. The displayed identity gives existence, since the first summand is symmetric and the second skew-symmetric. For a skew-symmetric matrix $\beta(x,x) = x^tAx$ equals its own transpose and also its own negative, hence vanishes, so skew-symmetric forms are alternating; conversely expanding $\beta(x+y, x+y) = 0$ gives $\beta(x,y) = -\beta(y,x)$. Uniqueness: if $S + N = S' + N'$ with S, S' symmetric and N, N' alternating then $S - S' = N' - N$ is both, hence zero. □

Theorem 22.14 (Alternating forms have one invariant). *Let β be alternating on V with $\dim V = n$. Then V has a basis in which the matrix of β is block diagonal with r copies of $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ followed by zeros. In particular the rank $2r$ is even, and it is the only invariant.*

Proof. Induct on n. If $\beta = 0$ take any basis. Otherwise choose u, v with $\beta(u, v) \neq 0$ and rescale v so that $\beta(u, v) = 1$; note u, v are independent, since $\beta(u, u) = 0$. Put

$$W \;=\; \{x \in V : \beta(u, x) = \beta(v, x) = 0\}.$$

Then $V = \operatorname{span}(u, v) \oplus W$. Indeed, for $x \in V$ set $x' = x - \beta(u, x)\, v + \beta(v, x)\, u$; a direct computation using $\beta(u, u) = \beta(v, v) = 0$ and $\beta(v, u) = -1$ gives $\beta(u, x') = \beta(v, x') = 0$, so $x' \in W$ and x lies in $W + \operatorname{span}(u, v)$. The sum is direct because $au + bv \in W$ forces $\beta(u, au + bv) = b = 0$ and $\beta(v, au + bv) = -a = 0$. The restriction of β to W is alternating, so induction applies, and in the basis u, v followed by a basis of W the matrix has the stated form. $\square$

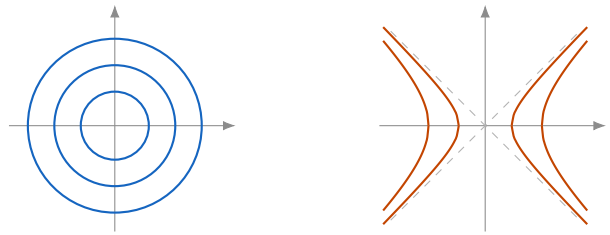
Key Idea

Compare the two classifications. A symmetric real form has the inertia (n_+, n_-, n_0) of Chapter 15 as its complete invariant, so there are $\binom{n+2}{2}$ of them up to congruence. An alternating form has only its rank, and that rank must be even, so there are $\lfloor n/2 \rfloor + 1$ of them. Symmetric forms carry geometry; alternating forms carry almost nothing, which is exactly why the nondegenerate one is the same for everybody and gets a name, the symplectic form, and a group of its own in Chapter 23.

Example 22.15 (Sesquilinear forms). *Over $\mathbb{C}$ the useful notion is* sesquilinear*: linear in one argument and conjugate-linear in the other, as the complex inner product of Definition 12.16 already was. Such a form has matrix $a_{ij} = \beta(v_i, v_j)$ with $\beta(x, y) = x^* A y$, it is called* Hermitian *when $A^* = A$, and the change of basis rule is $A \mapsto C^* A C$, called* conjugate congruence*. A Hermitian form takes only real values on the diagonal, since $\overline{x^* A x} = x^* A^* x = x^* A x$, and Sylvester's law of inertia holds verbatim with the same proof.*

22.5 Where the distinction is felt

Figure 22.4 shows what a congruence cannot do. Both level sets there belong to symmetric matrices, and any invertible change of coordinates may stretch, rotate and shear the pictures; what it cannot do is turn one into the other, because the signature is unchanged.



$x^2 + y^2$, signature $(2, 0)$ $x^2 - y^2$, signature $(1, 1)$

Figure 22.4. The signature is what survives a change of basis. Both forms above can be written as $x^t Ax$ for a symmetric A, and any invertible change of coordinates may stretch, rotate and shear the pictures; what it cannot do is turn one into the other. Sylvester's law of inertia says the pair of numbers, how many positive and how many negative squares appear after diagonalising, is an invariant of the form rather than of the basis. Geometrically it is the difference between closed level sets and unbounded ones, and no deformation closes a hyperbola. The degenerate case, a zero eigenvalue, gives a third count and the triple is complete.

Insight

A gradient is a functional and a step is a vector, and they live in different spaces. In coordinates both are lists of n numbers, so the difference is invisible until the coordinates change, at which point a vector's components transform by C^{-1} and a functional's by C^t; the physicists' contravariant and covariant indices of Chapter 20 are bookkeeping for exactly this. Gradient descent adds the gradient to the parameters, which is an addition of a functional to a vector and therefore not a well-defined operation until something converts one to the other. That something is an inner product, by Proposition 22.5. Plain gradient descent silently chooses the standard one; a preconditioner chooses a different one, and so does the natural gradient, and so does the Kronecker-factored curvature of Example 20.10. Seen this way those methods are not tricks. They are answers to a question the naive algorithm never asked.

Open Problem 22.16 (How many squares does a nonnegative polynomial need?)**.** *A quadratic form that takes only nonnegative values is a sum of squares of linear forms, and this chapter's diagonalisation says how many are needed: the rank. Ask the same question for rational functions. The* Pythagoras number $p(K)$ *of a field K is the smallest m such that every sum of squares in K is a sum of m squares.* What is the Pythagoras number of $\mathbb{R}(x, y, z)$?

What is known. *It is between* 5 *and* 8*, and that interval has not moved in fifty years. For* $\mathbb{R}(x)$ *the answer is* 2*, and for* $\mathbb{R}(x,y)$ *it is* 4*, a theorem of Cassels, Ellison and Pfister whose proof runs through elliptic curves and is one of the most surprising in the subject [137]. Pfister proved the general upper bound* $p(\mathbb{R}(x_1,\dots,x_n)) \le 2^n$ *[510], which gives* 8 *in three variables, and lower bounds of* 5 *are known from explicit forms.* Whether the answer is 5, 6, 7 or 8 is open.

Why it belongs with bilinear forms. *A sum of* m *squares is a quadratic form of rank* m*, so the question is precisely about the minimum rank in a family of positive semidefinite forms over a field that is not real closed. Pfister's bound comes from the theory of multiplicative forms, the descendants of the identity* $(a^2+b^2)(c^2+d^2) = (ac-bd)^2+(ad+bc)^2$ *that this chapter's readers will recognise as the norm form of the complex numbers, iterated.*

Where a reader might start. *Prove* $p(\mathbb{R}(x)) = 2$*: a sum of squares of rational functions in one variable is a nonnegative rational function, hence a quotient of nonnegative polynomials, and a nonnegative real polynomial in one variable factors as a product of squares and irreducible quadratics, each of which is a sum of two squares. Follow that argument and see exactly which step fails in two variables. Chapter 62 takes up Hilbert's seventeenth problem, which this question is a quantitative refinement of.*

Exercises

Exercise 22.17. *Let* $V = \mathbb{R}^2$ *with basis* $v_1 = (1,1)$*,* $v_2 = (1,-1)$*. Write the dual basis* f_1, f_2 *explicitly as functions of* (x,y)*, and check* $f_i(v_j) = \delta_{ij}$*. Then verify* $v = f_1(v)v_1 + f_2(v)v_2$ *for* $v = (3,5)$*.*

Exercise 22.18. *Show that the isomorphism* $V \to V^*$ *of Proposition 22.3 really does depend on the basis, by computing where* $v_1 = (1,0)$ *goes under the standard basis of* $\mathbb{R}^2$ *and under the basis* $(1,0),(1,1)$*, and observing that the two functionals differ.*

Exercise 22.19. *Show that* $X \mapsto \operatorname{trace}(AX)$ *is a linear functional on* $M_n(\mathbb{R})$ *for each fixed* A*, and that every functional on* $M_n(\mathbb{R})$ *arises this way from exactly one* A*. Which inner product does this exhibit as the Riesz identification?*

Exercise 22.20. *Let* $T : \mathbb{R}^3 \to \mathbb{R}^2$ *be* $T(x,y,z) = (x+y,\ y-z)$ *and let* $g \in (\mathbb{R}^2)^*$ *be* $g(s,t) = 2s - t$*. Compute* $T^t g$ *directly from Definition 22.6, then again from Theorem 22.7,*

and check that the answers agree.

Exercise 22.21. *Let $U \subset \mathbb{R}^4$ be spanned by $(1,0,1,0)$ and $(0,1,0,1)$. Find a basis of U° and check Proposition 22.9. Then find $(U^\circ)^\circ$ under the identification of Theorem 22.4 and say what you notice.*

Exercise 22.22. *Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1/2 \end{smallmatrix}\right)$. Compute $C^{-1}AC$ and C^tAC and compare their eigenvalues. Which quantities survive each operation, and which of the two is the right one if A is the matrix of a quadratic form?*

Exercise 22.23. *Write down the matrix of the form $\beta(x,y) = x_1y_2 - x_2y_1 + x_3y_4 - x_4y_3$ on $\mathbb{R}^4$, verify that it is alternating, and identify r in Theorem 22.14. Then show directly that a skew-symmetric matrix of odd size is singular, and give two proofs, one using Theorem 22.14 and one using Proposition 8.13.*

Exercise 22.24. *Decompose $A = \left(\begin{smallmatrix} 1 & 4 \\ 0 & 1 \end{smallmatrix}\right)$ into its symmetric and alternating parts as in Proposition 22.13. Which part does the quadratic form $x \mapsto x^tAx$ see, and why does that make the alternating part invisible to Chapter 15?*

Exercise 22.25. *Give the proof of $\operatorname{rank} A = \operatorname{rank} A^t$ that follows from Theorem 22.10 in your own words, and say which of the three proofs in this book works over an arbitrary field and which needs $\mathbb{R}$ or $\mathbb{C}$.*

References and further reading

Halmos [295] is the classic short treatment of everything in this chapter and takes duality seriously from the start, which most first courses do not; his account of annihilators is the one followed here. Axler [37] devotes a careful chapter to duality in the fourth edition and is good on why the double dual identification deserves the word canonical. For bilinear and sesquilinear forms and their classification over general fields, including the cases $\mathbb{F}$ finite or of characteristic two where Proposition 22.13 fails, Artin [28] and Lang [399] are the standard algebraic references, and Artin is also where the symplectic group of Chapter 23 appears in its natural setting. Treil [610] is the source of the remark in Chapter 7 that this chapter has now answered. Banica [51], our baseline, treats forms

and signature as the entry point to its part on positive matrices, which is the route Chapter 15 took.